Roger Tawfik

(508) 988-0728 | Natick, MA | tawfikroger@gmail.com | rogertawfik.github.io | LinkedIn | GitHub

Education

University of Massachusetts Amherst — **Amherst, MA**
Bachelor of Science in Computer Engineering — *Expected May 2028*

- **Academic Distinctions:** GPA: 3.804/4.00 | Dean's List (Fall 2024, Spring 2025, Fall 2025, Spring 2026)

Technical Skills

Languages: C, C++, Python, TypeScript, Java, Kotlin, Swift, Assembly (ARM/AVR)
ML, AI & Computer Vision: OpenCV, MediaPipe Tasks, TFLite, Core ML, MeTRAbs, YOLOv8, LangChain, LightGBM, Random Forest
Full-Stack: React, Vite/PWA, Supabase (Auth/Postgres/Edge Functions), REST API integration
Embedded Systems: Digital Logic, Circuits & Electronics, Arduino firmware development, Concurrency
Core CS: Data Structures, Algorithms, Object-Oriented Programming
Data & Tools: Pandas, NumPy, Git, Linux, GCC, MATLAB, SQLite, LM Studio

Work Experience

I-SENSE, Florida Atlantic University — **Jun 2026 - Present**
NSF REU Researcher (Undergraduate Research Intern) — *Boca Raton, FL*
Python, Kotlin, Swift, OpenCV, MediaPipe Tasks, YOLOv8, TFLite, Core ML, LangChain, MeTRAbs, Git
Health & Behavior: Wearables & Data Analytics for Personalized Therapeutic Management, advised by Dr. Behnaz Ghoraani [Project Page]

- Built an end-to-end clinical gait video QC pipeline across Python, native Android, and native iOS to evaluate recording quality, extract body-joint data, and support automated gait-metric computation
- Developed a tiered, fail-fast QC framework combining YOLOv8 person tracking with MediaPipe pose estimation to detect lighting, focus, occlusion, framing, and walking-orientation issues before downstream analysis
- Built dual-path LLM reporting with LangChain for offline local inference and a cloud-hosted model path, generating technical diagnostics for developers and plain-language guidance for clinicians with structured logs and diagnostic visualizations
- Implemented native Kotlin and Swift versions of the Python pipeline, reproducing consistent Android and iOS verdicts on desktop reference clips and documenting platform-specific model behavior

Personal Projects

AI Nutrition Tracker App | *TypeScript, React, Supabase, Vite/PWA* — Jul 2026 - Present

- Built a nutrition tracker for natural language, photo, and barcode food logging, with adaptive calorie targets, gap-fill nutrient recommendations, and daily AI coaching insights
- Architected as a mobile-first PWA with Supabase auth, Postgres, and edge functions, using a review-before-log workflow to keep AI-parsed entries accurate

Gym Form Trainer | *Python, C, OpenCV, MediaPipe, Arduino* — Dec 2025 - May 2026

- Built a real-time computer vision fitness trainer in Python using MediaPipe Pose and OpenCV for full-body landmark extraction, HUD rendering, and color-coded form scoring
- Designed and flashed C firmware to an Arduino microcontroller for live LED/buzzer feedback via bidirectional serial communication, closing the loop from sensor input to physical hardware output

Algorithmic Trading Analysis System | *Python, Pine Script, SQLite, LightGBM* — Feb 2026 - Apr 2026

- Engineered an end-to-end automated market analysis system in Python to screen 5,000+ stocks using multi-factor technical signals and quantitative scoring models for breakout detection
- Built a trigger-aware historical labeling pipeline over 660,000+ ticker-date observations, using LightGBM, time-ordered validation, SHAP analysis, isotonic calibration, SQLite caching, and TradingView Pine Script backtesting

Service Mesh Graph Project | *Python* — Nov 2025 - Dec 2025

- Designed and implemented a weighted graph-based service mesh simulation in Python using OOP and modularity
- Modeled services as vertices and latency links as edges, using priority queues for routing and path optimization
- Built BFS reachability, Dijkstra shortest path, and Prim MST analyses for connectivity and optimization